BUREAU OF INDIAN STANDARDS

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भारतीय मानक मसौदा

चमड़े की भौतिक परीक्षण की पद्धतियाँ

भाग 5 विदारण भार का निर्धारण

अनुभाग 2 डबल एज विदारण

(*पहला पुनरीक्षण*)

{*IS 5914 (Part 5/Sec 2) Identical adoption under dual numbering of ISO 3377-2 : 2025*}

Draft Indian Standard

Methods of Physical Testing of Leather

Part 5 Determination of Tear Load

Section 2 Double Edge

{*IS 5914 (Part 5/Sec 2) Identical adoption under dual numbering of ISO 3377-2 : 2025*}

(*First Revision*)

(ICS 59.140.30)

Leather, Tanning Materials and Allied Products, Sectional Committee, CHD 17	Last Date for Comments: **As per the covering letter**

NATIONAL FOREWORD

(*Formal clause will be added later*)

The Indian Standard IS 5914 : 1970 Methods of physical testing of leather prescribes the methods for carrying out physical tests for all types of leathers. The Committee responsible for formulating this standard has decided to harmonize the methods of test prescribed in IS 5914 with those prescribed in ISO/IULTCS standards. Accordingly, the Committee decided to retain IS 5914 and publish the harmonized/adopted test methods published by ISO/IULTCS in various parts of IS 5914 as this standard is widely recognized by the Indian Leather Industry.

The Committee further decided to publish the adopted/harmonized standards in the following manner:

a) Wherever an existing test method is being replaced by the corresponding ISO/IULTCS test method, the relevant part will be published as revision with the information in the national foreword about the method of IS 5914 being superseded; and
b) When a new test method is being incorporated in IS 5914 the same will be published as a new standard and as subsequent part of IS 5914.

This Indian standard was first published in 2023 as an identical adoption of ISO 3377-2 : 2016 under dual numbering. ISO 3377-2 : 2016 was revised and published as ISO 3377-2 : 2025. The first revision of IS 5914 (Part 5/Sec 2) has been brought out to adopt the revised version ISO 3377-2 : 2025. In this revision, following modifications have been incorporated:

a) Alignment with the changes and revised terminology in IS 5868 (Part 2) : 2025/ ISO 2418 and IS 5868 (Part 3) : 2026/ISO 2419; and
b) Addition of Clause 3.

This (Part 5/Sec 2) specifies a method for determining the tear strength of leather using a double edged tear. The method is sometimes described as the Baumann tear. It is applicable to all types of leather.

IS 5914 have been published in several parts. The other part of this series are:

Part 1 Determination of water vapour absorption

Part 2 Determination of abrasion resistance

Sec 1 Taber method

Sec 2 Martindale ball plate method

Part 3 Determination of soiling

Sec 1 Rubbing (Martindale) method

Sec 2 Tumbling method

Part 4 Determination of apparent density and mass per unit area

Part 5 Determination of tear load

Sec 1 Single edge tear

Sec 2 Double edge tear

Part 6 Determination of flex resistance

Sec 1 Flexometer method

Sec 2 Vamp flex method

Part 7 Determination of resistance to grain cracking and grain crack index

Part 8 Determination of tensile strength and percentage elongation

Part 9 Determination of heat resistance of patent leather

Part 10 Determination of surface coating thickness

Part 11 Determination of thickness

Part 12 Determination of water vapour permeability

Part 13 Determination of softness

Part 14 Determination of resistance to horizontal spread of flame (under publication Doc No. : 28698)

Part 15 Determination of water resistance of heavy leathers (under publication Doc No. : 28699)

Part 16 Leather — Determination of water resistance of flexible leather

Sec 1 Repeated linear compression (Penetrometer) (under publication Doc No. : 28714)

Part 17 Determination of water resistance repellency of garment leather (under publication Doc No. : 28715)

Part 18 Determination of the static absorption of water (under publication Doc No. : 34133)

The text of ISO Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions and terminologies are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

a) Wherever the words `International Standard' appear referring to this standard, they should be read as `Indian Standard'
b) Comma (,) has been used as a decimal marker in the International Standard, while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards, which are to be substituted in their places, are listed below along with their degree of equivalence for the editions indicated:

International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
ISO 2418 Leather — Chemical, physical, mechanical and fastness tests — position and preparation of specimens for tasting	IS 5868 (Part 2): 2025/ISO 2418: 2023 Leather — Methods of Sampling Part 2 Positionand preparation of specimens for testing for chemical, physical, mechanical and fastness tests (*second revision*)	Identical
ISO 2419 Leather — Physical and mechanical tests — Specimen and test piece conditioning	IS 5868 (Part 3):2026/ ISO 2419 : 2024 Leather — Methods of Sampling Part 3 Specimen and Test Piece Conditioning for Physical and Mechanical Tests (*second revision)*	Identical
ISO 2589 Leather — physical and mechanical test — Determination of thickness	IS 5914 (Part 11):2025/ ISO 2589 : 2016 Methods of Physical Testing of Leather Part 11 Determination of Thickness	Identical
ISO 7500-1 Metallic materials — Calibration and verification of static uniaxial testing machines — Part 1: Tension/compression testing machines — Calibration and verification of the force — measuring system	IS 1828 (Part 1):2022/ ISO 7500-1 : 2018 Metallic Materials - Calibration and Verification of Static Uniaxial Testing Machines - Part 1: Tension/Compression Testing Machines - Calibration And Verification of the Force-Measuring System	Identical

In this adopted standard, reference appears to certain International Standards where the standard atmospheric conditions to be observed are stipulated which are not applicable to tropical/subtropical countries. The applicable standard atmospheric conditions for Indian conditions are 27 °C ± 2 °C and (65 ± 5) percent relative humidity and shall be observed while using this standard.

In reporting the result of a test or analysis made in accordance with this standard, if the final value, observed or calculated, is to be rounded off, it shall be done in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'.

FOR COMPLETE TEXT OF THE DOCUMENT, KINDLY REFER ISO 3377-1 : 2025

Note: The technical content of the document has not been enclosed as these are identical with the corresponding ISO Standard. For obtaining the copy of the complete ISO Standard, please contact:

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